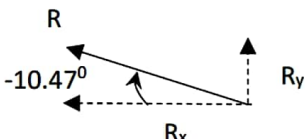
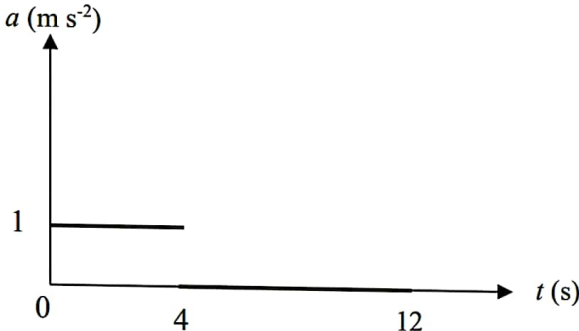
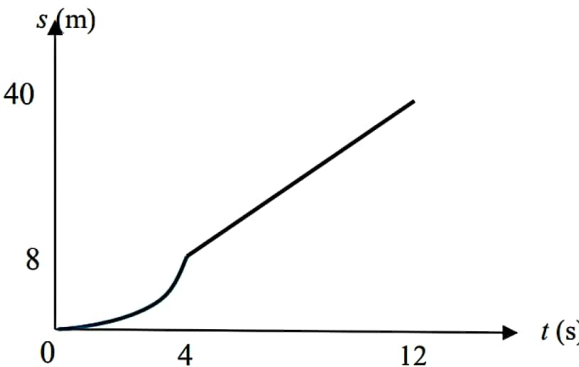


**SKEMA JAWAPAN & PEMARKAHAN**

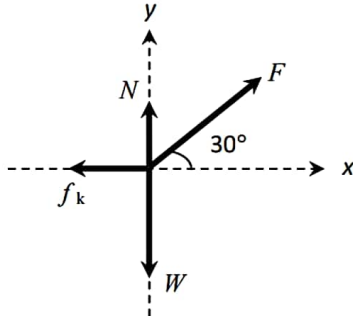
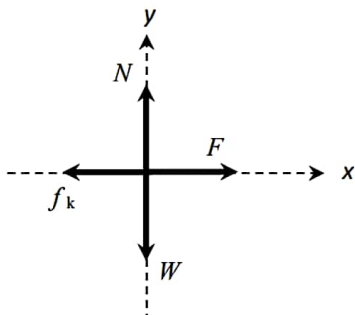
No	Scheme												
1 (a)	$m = 571.5 \text{ g}$ $= 571.5 \times 10^{-3} \text{ kg}$ $V = 2250 \text{ cm}^3$ $= 2250 \times 10^{-6} \text{ m}^3$ $\rho = \frac{m}{V} = \frac{571.5 \times 10^{-3}}{2250 \times 10^{-6}} = 254 \text{ kg m}^{-3}$												
1 (b)	<table><tr><th>vector</th><th>x - component</th><th>y - component</th></tr><tr><td>A</td><td>$A_x = -A \sin 50^\circ$ $= -6 \sin 50^\circ$ $= -4.60 \text{ m}$</td><td>$A_y = A \cos 50^\circ$ $= 6 \cos 50^\circ$ $= 3.86 \text{ m}$</td></tr><tr><td>B</td><td>$B_x = -B \cos 70^\circ$ $= -3 \cos 70^\circ$ $= -1.03 \text{ m}$</td><td>$B_y = -A \sin 70^\circ$ $= -3 \sin 70^\circ$ $= -2.82 \text{ m}$</td></tr><tr><td>R</td><td>$R_x = A_x + B_x$ $= -4.60 + (-1.03)$ $= -5.63 \text{ m}$</td><td>$R_y = A_y + B_y$ $= 3.86 + (-2.82)$ $= 1.04 \text{ m}$</td></tr></table> $R = \sqrt{R_x^2 + R_y^2}$ $= \sqrt{(-5.63)^2 + (1.04)^2}$ $= 5.73 \text{ m}$ $\theta = \tan^{-1} \left(\frac{R_y}{R_x} \right)$ $\theta = \tan^{-1} \left(\frac{-9.39}{0.61} \right)$ $= -10.47^\circ$ Above negative x-axis. 	vector	x - component	y - component	A	$A_x = -A \sin 50^\circ$ $= -6 \sin 50^\circ$ $= -4.60 \text{ m}$	$A_y = A \cos 50^\circ$ $= 6 \cos 50^\circ$ $= 3.86 \text{ m}$	B	$B_x = -B \cos 70^\circ$ $= -3 \cos 70^\circ$ $= -1.03 \text{ m}$	$B_y = -A \sin 70^\circ$ $= -3 \sin 70^\circ$ $= -2.82 \text{ m}$	R	$R_x = A_x + B_x$ $= -4.60 + (-1.03)$ $= -5.63 \text{ m}$	$R_y = A_y + B_y$ $= 3.86 + (-2.82)$ $= 1.04 \text{ m}$
vector	x - component	y - component											
A	$A_x = -A \sin 50^\circ$ $= -6 \sin 50^\circ$ $= -4.60 \text{ m}$	$A_y = A \cos 50^\circ$ $= 6 \cos 50^\circ$ $= 3.86 \text{ m}$											
B	$B_x = -B \cos 70^\circ$ $= -3 \cos 70^\circ$ $= -1.03 \text{ m}$	$B_y = -A \sin 70^\circ$ $= -3 \sin 70^\circ$ $= -2.82 \text{ m}$											
R	$R_x = A_x + B_x$ $= -4.60 + (-1.03)$ $= -5.63 \text{ m}$	$R_y = A_y + B_y$ $= 3.86 + (-2.82)$ $= 1.04 \text{ m}$											

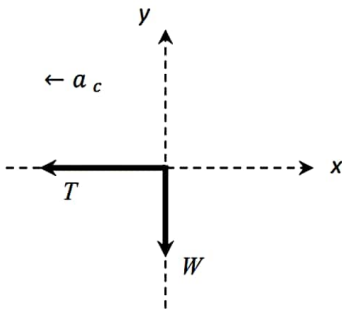


No	Scheme
2 (a) (i)	Acceleration = gradient of graph v-t For $t = 0$ s to $t = 4$ s, $a_{0-4} = \frac{4 - 0}{4 - 0} = 1 \text{ m s}^{-2}$ For $t = 4$ s to $t = 12$ s, $a_{4-12} = \frac{4 - 4}{12 - 4} = 0 \text{ m s}^{-2}$ 
2 (a) (ii)	Displacement = area under graph v-t $s_{0-4} = \frac{1}{2}(4)(4) = 8 \text{ m}$ $s_{4-12} = (4)(12 - 4) = 32 \text{ m}$ 
2 (b) (i)	$s = ut - \frac{1}{2}gt^2$ $= (5)(1.2) - \frac{1}{2}(9.81)(1.2)^2$ $= -1.06 \text{ m}$
2 (b) (ii)	$v = u - gt$ $= (5) - (9.81)(1.2)$ $= -6.77 \text{ m s}^{-1}$

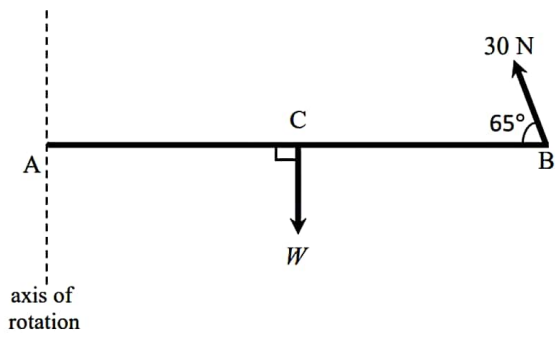
No	Scheme
3 (a)	$\sum p_i = \sum p_f$ $m_1 u_1 + m_2 u_2 = (m_1 + m_2)v$ $(0.4)(+15) + (0.6)(-15) = (0.4 + 0.6)v$ $v = -3 \text{ m s}^{-1}$
3 (b)	$\sum p_i = \sum p_f$ $m_1 u_1 + m_2 u_2 = m_1 v_1 + m_2 v_2$ $(2)(+5) + (4)(-8) = (2)v_1 + (4)(+4)$ $v = -19 \text{ m s}^{-1}$
3 (c) (i)	$J = F \Delta t$ $= (8)(1 \times 60)$ $= 480 \text{ kg m s}^{-1}$
3 (c) (ii)	$J = m(v - u)$ $480 = 18(60 - u)$ $u = 33.33 \text{ m s}^{-1}$

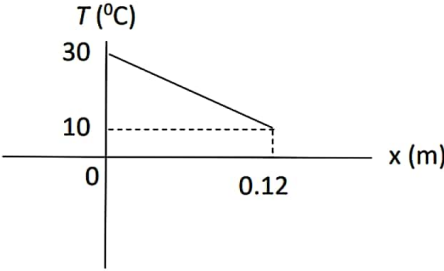


No	Scheme
4 (a) (i)	Free body diagram: 
4 (a) (ii)	$\sum F_y = 0$ $N + F \sin \theta = W$ $N + (400) \sin 30^\circ = (70)(9.81)$ $N = 486.7 \text{ N}$
4 (a) (iii)	$f_k = \mu_k N$ $= (0.50)(486.7)$ $= 243.35 \text{ N}$
4 (a) (iv)	$\sum F_x = ma$ $F \cos \theta - f = ma$ $(400) \cos 30^\circ - (243.35) = (70)a$ $a = 1.47 \text{ m s}^{-2}$
4 (b) (i)	Free body diagram:  $\sum F_y = 0$ $N = W$ $N = (60)(9.81)$ $N = 588.6 \text{ N}$ $\sum F_x = 0$ $f_k = F$ $\mu_k (588.6) = 140$ $\mu_k = 0.238$

No	Scheme
5 (a) (i)	$v = r\omega$ $(0.3) = (0.5)\omega$ $\omega = 0.6 \text{ rad s}^{-1}$
5 (a) (ii)	$\theta = \omega t$ $\theta = (0.6)(3 \times 60)$ $= 108 \text{ rad}$ $= \frac{108}{2\pi} = 17.19 \text{ revolutions}$
5 (b) (i)	Free body diagram: 
5 (b) (ii)	$\sum F_x = F_c$ $T = m \frac{v^2}{r}$ $(50) = (0.2) \frac{v^2}{(1.5)}$ $v = 19.36 \text{ m s}^{-1}$



No	Scheme
6 (a) (i)	$v = u + at$ $(16) = (0) - a(8)$ $a = 2 \text{ m s}^{-2}$
6 (a) (ii)	$r = \frac{d}{2} = \frac{0.5}{2} = 0.25 \text{ m}$ $a = r\alpha$ $(2) = (0.25)\alpha$ $\alpha = 8 \text{ rad s}^{-2}$
6 (a) (iii)	$\theta = \omega_0 t + \frac{1}{2} \alpha t^2$ $= 0(8) + \frac{1}{2} (8)(8)^2$ $= 256 \text{ rad}$
6 (a) (iv)	$\theta = \frac{256}{2\pi} = 40.74 \text{ revolutions}$
6 (b)	Free body diagram:  axis of rotation $\tau_C = r_C F_C \sin \theta_C$ $\tau_C = (2)[(5)(9.81)] \sin(90^\circ)$ $= -98.1 \text{ N m}$ $\tau_B = r_B F_B \sin \theta_B$ $\tau_C = (4)(30) \sin(65^\circ)$ $= +108.76 \text{ N m}$ $\sum \tau = \tau_C + \tau_B$ $= (-98.1) + (+108.76)$ $= +10.66 \text{ N m}$

No	Scheme
7 (a) (i)	$\frac{Q}{t} = -kA \frac{dT}{dx}$ $= -(0.84)(4 \times 4) \left(\frac{10 - 30}{0.12} \right)$ $= 2240 \text{ W}$ The number of light bulbs $= \frac{2240}{112} = 20$
7 (a) (ii)	
7 (b) (i)	$\frac{m}{M} = \frac{N}{N_A}$ $\frac{m}{0.028} = \frac{1}{3.35 \times 10^{-27}}$ $m = 4.65 \times 10^{-26} \text{ kg}$
7 (b) (ii)	$v_{rms} = \sqrt{\frac{3RT}{M}}$ $= \sqrt{\frac{(3)(8.31)(0 + 273.15)}{0.028}}$ $= 493.14 \text{ m s}^{-1}$
7 (b) (iii)	$v_{rms} \propto \sqrt{T}$ $\frac{(v_{rms})_2}{(\sqrt{T})_2} = \frac{(v_{rms})_1}{(\sqrt{T})_1}$ $T_2 = \left(\frac{2(v_{rms})_1}{(v_{rms})_1} \right)^2 (273.15)$ $= 1092.6 \text{ K}$
7 (c) (i)	$\Delta U = Q - W$ $\Delta U = (0) - (+5)$ $\Delta U = -5 \text{ J}$
7 (c) (ii)	$\Delta U = Q - W$ $(0) = Q - (-8)$ $Q = -8 \text{ J}$



